Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Maxwell Hankey	mhankey@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

Material selector/Comparison.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There are already many programs to see how the materials do under stress/load and what their properties are.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

Input how much money you have, how big your project is and how much stress the objects will be under and it will spit out the best material for you based on what you inputted into the program

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

To take in specifications about a build you need and it will give you what material makes the most sense to be used

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Amount of money they wish to spend, volume of material they need, how much stress the material would be under(is it gonna be used for a car or a fence), does the weight matter, location(heat purposes),

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

It will check each one by themselves and determine which material works the best then which ever material shows up the most from what they inputted will be given back to them

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The material that shows up the most based off of the constraints

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Location, weight, volume, cost, stress,

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

All the them I need to worry about

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

Creating the location dataclass because it will most likely be a dataclass inside of a dataclass

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I probably will not be using LLM unless I get majorly stuck after asking for help from office hours

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

Learn drafter better/learn more indepth python strategies for approaching problems

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

What are more ideas that are asked for in the prompt from the user

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

Adding a screen to see all the the users past responses to allow them to re visit what they had previously asked for

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

Making it a list of what they previously said

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

More focused on a specific project

Part 6: Resource Planning

1. What resources will you use if you get stuck?

Professors/Office Hours then LLM if necessary

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

Remove what the LLM said and run it again then try to reverse engineer with topics we have used before.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

None

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.

Final Project Design Phase

https://mhankey-cyber.github.io/Material-Finder/

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Loops and lists will be used basically the entire time. I will probably have at least one dataclass for the material. I definitely want to get better at the syntax for drafter and for setting up routes and pages

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

Everything other than the implementation to the drafter. Most of the work we have done in class has been very easy when it's one basic concept but drafter I haven't had a lot of time to work with so it will most likely be the hardest part for me to complete.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

Creating all the different dataclass instances for the different materials. It will not be

very hard, just very time consuming but it will have to be the first thing that I do before implementing any functions.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

I have no questions about the assignment and the only help I should hopefully need throughout the assignment is help implementing the drafter routes and pages. Help with the syntax. I know the basics but some of the smaller stuff I might need help on.

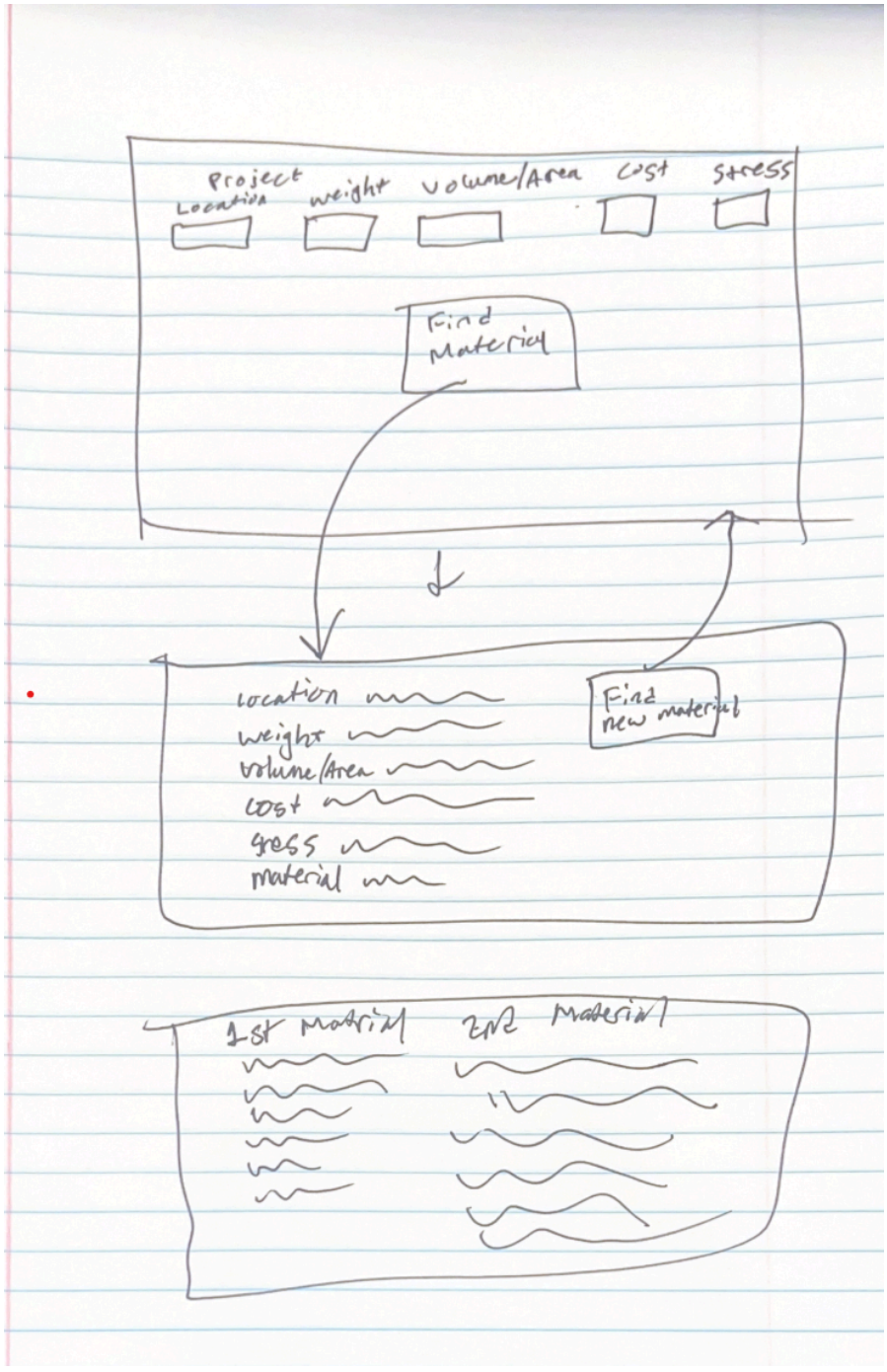
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Project Location
weight
volume/Area
cost
stress
Find Material
location
weight
volume/Area
cost
gross
material
Find new material
1st Material
2nd Material

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	Asks for 5 different categories of location, weight, cost, Volume/Area, stress and gives a material that best fits your constraints. It displays all your previous constraints and the chosen material on a screen and then if you use it again the screen displays all of your past cases and current cases.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	State: Object constraints Weight of Object: float Total Cost: int Stress on Object: int Volume of Material: float Location: Location dataclass Possible Materials: list[material dataclass]
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Location dataclass Average temperature: int Weather: str Material Dataclass Density: float Cost per pound: str Is water resistant: bool Force to bend or break: int The material dataclass should have a set of 20 instances for each of the different materials the

	program could return
Routes (minimum 7) The names of your route functions	-Find material: takes in a state and returns a screen with your constraints and the material that works the best based on your constraints. -Change Weight: allows you to change the weight and see if you would get a different material by just changing the weight. -Change cost: allows you to change the cost and see if you would get a different material by just changing the cost. -Change location: allows you to change the location and see if you would get a different material by just changing the location. -Change Volume: allows you to change the volume and see if you would get a different material by just changing the volume. -Change Stress: allows you to change the stress and see if you would get a different material by just changing the stress. -Add new constraints: allows the user to change all the constraints again like they did when first picking the restraints.
Pages (minimum 5) A brief description of the pages in your site	Home Page: displays buttons to get to the instruction page, list of material page, and to input object constraint page. Instructions Page: has instructions on how the user needs to input certain information and how the website works Object Constraint Page: where the user can input their first set of constraints. List of All possible Material Page: list of all the material data classes that the website could return Screen of your constraints and your chosen material Page: where the user can see their chosen material and where they can change any of their constraints to get a different material.
User Input (minimum 3)	Takes in all the object constraints which are 5 things

Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	the user inputs then the user can change what they input later if they want to
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	To check if a material could fit in their constraints One if statement per restraint
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	To loop through each one of the materials in the state to check which one works the best.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Replace this text with your answer.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.